

Visualizing Spatial and Temporal Patterns in NYC Airbnb Performance Using an Interactive Multi-View Dashboard

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Abstract

This project presents an interactive web-based visualization system for exploring monthly Airbnb market dynamics in New York City. Using a coordinated multi-view dashboard, the system combines spatial, temporal, and categorical views to support analysis of revenue, occupancy, pricing, and room-type differences. The dashboard was implemented with a Python backend and a custom web frontend to enable interactive filtering and comparison across ZIP codes and time periods. The results show that Airbnb revenue is spatially concentrated and varies substantially across both time and room type, demonstrating the value of interactive visualization for urban rental market analysis.

1 Introduction and Motivation

Short-term rental markets such as Airbnb are shaped by the interaction of location, price, demand, and time. In a dense urban environment like New York City, hosts and analysts often need to compare not only how listings perform across neighborhoods, but also how market conditions change from month to month. Traditional descriptive summaries such as tables, static averages, and one-off charts are limited because they flatten spatial and temporal variation into a few aggregate statistics. As a result, it becomes difficult to answer questions such as: Which areas show consistently strong revenue performance? How do room types differ in their pricing and occupancy behavior? Are high prices associated with high revenue, or are some locations overpriced relative to demand?

This project addresses that problem by building an interactive web-based visualization system for exploring monthly Airbnb performance in New York City. The goal is not simply to display the data, but to support visual analysis through coordinated views that combine map-based, temporal, categorical, and multivariate perspectives. The system is intended for prospective Airbnb hosts, small-scale property managers, and students or analysts interested in urban short-term rental dynamics. For these users, the dashboard provides a practical way to inspect area-level market conditions, compare room-type segments, and detect trends that would be difficult to identify from raw spreadsheets alone.

2 Related Work

The related work for this project can be classified into two main categories, visualization design research and interactive visualization tools. In design, the study of legibility and visual perception (Ware, 2004; Munzner, 2014) shows that no single representation can be used for all analytical tasks, and that successful visualization depends on matching visual form to the structure of the data and the questions the user wants to answer. For example, studies comparing alternative visual representations show that different views support different forms of comparison and pattern recognition, which is a direct motivation for this project’s use of multiple coordinated views rather than a single chart. In addition, color perception and visual aesthetics studies highlight the need for perceptually meaningful encodings, a legible hierarchy and purposeful color differentiation. These ideas are especially relevant for the current dashboard because the dataset combines spatial, temporal, categorical, and quantitative attributes, making it easy to overload a single display.

On the implementation side, tools such as Plotly and Leaflet provide the practical foundation for building interactive analytical systems (Heer and Shneiderman, 2012), since they support dynamic chart updates, hover interaction, map navigation, and multi-view exploration in a web environment. However, these tools by themselves do not solve the analytical problem; they only provide general components. Compared with prior work, this project contributes a domain-specific visual analytics system for exploring NYC Airbnb performance across ZIP codes, room types, and time. Existing visualization research provides general principles for readability, perception, and interaction, while general-purpose libraries provide technical capability, but neither directly addresses the challenge of bringing spatial, temporal, and listing-level Airbnb performance into one coherent interface.

We do not propose a new visualization technique. Instead, the niche this project fills is the design and implementation of an interpretable, coordinated dashboard that applies established visualization principles to a real-world urban rental market dataset, and allows users to explore pricing patterns, occupancy variation, revenue concentration, and seasonal change in ways that are difficult to achieve through static summaries or isolated charts.

3 Data Description

The primary dataset used in this project is Airbnb Data: New York City 5Y (2021–2026) from Kaggle. The working system uses monthly Airbnb listing performance records from March 2021 to February 2026. The cleaned data set used by the dashboard after preprocessing contains 35,157 monthly observations, 975 unique listings, 872 unique hosts, and 161 ZIP code areas. The key variables used in the visualization include the geographical coordinates (latitude, longitude), temporal data (month_date), performance measures (avg_daily_rate, occupancy, revenue, and revpar), and categorical features like room_type, and property_type. These variables make the dataset suitable for a multi-view visual analysis system, as it combines spatial, temporal, categorical and quantitative information in the same source. But this complexity also makes raw inspection dif-

difficult, as meaningful interpretation requires comparison of multiple variables at multiple levels of aggregation.

The dataset required several preprocessing steps before it could be used in the final system. First, dates and numeric fields were parsed and standardized so that the records could be filtered consistently by month and used for quantitative comparison. The raw neighborhood field was also normalized into a `zip_code` field, since in practice it behaved more like a ZIP code identifier than a human-readable neighborhood name. Records with unresolved ZIP codes were removed from the processed dataset to avoid ambiguity in the map. In addition, boolean variables such as `active` and `superhost` were normalized, and several aggregated outputs were created for faster interaction in the web interface. A major challenge in the data is that many monthly observations contain `revenue = 0` and `occupancy = 0`, which usually indicates that a listing had no bookings in that month. There are also substantial missing values in variables such as `star_rating`, `bedrooms`, and booking-related attributes. Finally, because the data is monthly rather than daily, the dashboard cannot capture finer short-term fluctuations in demand or pricing. These limitations influenced both the cleaning process and the final design choices, especially the need for filtering, aggregation, and careful interpretation of zero-revenue records.

4 Design and Methodology

The system was conceived as a coordinated multi-view dashboard, integrating a spatial map, a monthly trend chart, a room-type comparison chart, and a listing-level scatter plot. We chose this layout to represent the structure of the dataset, which has geographic, temporal, categorical and quantitative variables that cannot be well visualized in one chart. The main view used was the map of the spatial distribution as location is the focal point of Airbnb market analysis, the time-series chart supports trend and seasonality analysis, the room-type comparison chart supports categorical comparison and the scatter plot reveals relationships between price, occupancy and revenue. These choices were guided by course concepts in visual encoding and perception: position is used for the most important quantitative comparisons, color is used as a supporting channel rather than the sole carrier of meaning, and marker size is treated as a secondary cue rather than a precise encoding. The interface also emphasizes clear hierarchy and readability by giving the spatial view the largest visual area and placing supporting views below it. Interaction was incorporated through date-range filtering, multi-select category filters, ZIP selection, map zooming, and coordinated updates across views, allowing users to explore subsets of the data without losing overall context. During development, alternative approaches such as a static dashboard, a single-view map, and a single-month map snapshot were considered, but the final design favored a range-based coordinated layout because it provided a better balance between accuracy, interpretability, and exploratory flexibility. Table 1 summarizes the main encoding decisions used in the dashboard and their justification in terms of visualization principles.

Data Variable	Data Type	Mark	Channel	Principle	Justification
ZIP code location	Spatial	Point	x/y position	Expressiveness	Geographic coordinates must be preserved spatially to support area comparison.
Revenue	Quantitative	Point	Color intensity	Effectiveness	Color provides a clear secondary cue for comparing area-level magnitude on the map.
Listing count	Quantitative	Point	Size	Expressiveness	Marker size indicates approximate market presence without replacing the primary metric.
Month	Temporal	Line	x-position	Ordering	Horizontal position makes temporal sequence easy to read.
Average revenue by month	Quantitative	Line	y-position	Effectiveness	Vertical position supports accurate comparison over time.
Room type	Categorical	Bar	x-position	Identity	Discrete position separates room-type categories clearly.
Average revenue by room type	Quantitative	Bar	y-position	Effectiveness	Bar height supports easy comparison across categories.
Average daily rate	Quantitative	Point	x-position	Effectiveness	Position on a common axis is effective for comparing prices.
Occupancy	Quantitative	Point	y-position	Effectiveness	Vertical position supports comparison of occupancy levels.
Revenue (listing-level)	Quantitative	Point	Size, color	Redundant encoding	Bubble size and color together highlight higher-performing listings in the scatter plot.

Table 1: Channel justification for the main views in the dashboard.

5 System Description and Implementation

The final system was implemented as a Python-based interactive web application with a modular frontend and backend structure. Also preprocessing and application logic were done in Python. A preprocessing script was used to clean the raw dataset from Airbnb, which parses the monthly dates, converts numeric and boolean fields, normalizes the raw neighborhood field into zip_code, removes unresolved ZIP records, and creates processed files for the dashboard. We chose Flask as our backend framework because it is lightweight and can easily serve the main dashboard page as well as multiple JSON API endpoints. These endpoints return filtered data for KPI summary, spatial view, monthly trend chart, room-type comparison chart, listing-level scatter plot. Separating preprocessing, server-side filtering and rendering helped to make the system more maintainable and reproducible.

HTML and CSS were used on the frontend to set the dashboard layout and visual hierarchy, and JavaScript was used to handle filter state and dynamically update all views. For the time-series chart, room-type comparison chart and scatter plot we used Plotly.js. Plotly.js is an efficient way to

support interactive statistical graphics. For the map showing the spatial distribution, Leaflet was used since it gives a more natural map interface, with zooming, panning and point-based popups. These tools have been integrated in this project to enable coordinated exploration across space, time and category. The key features of the final implementation are date range filter, multi-select controls for room type, property type and ZIP code, kpi summary cards, interactive ZIP level map and dynamic updates across all charts. Together, these components form a working visual analytics system rather than a collection of static charts.

The final dashboard was also deployed at <https://nyc-airbnb-market-explorer.vercel.app/>, allowing lightweight browser-based testing outside the local development environment.

6 Results and Insights

The interactive dashboard revealed two main patterns in the NYC Airbnb market. First, Airbnb revenue is not evenly distributed across space, but instead concentrated in a limited subset of ZIP codes. Second, market performance varies over time and also differs substantially across room types, suggesting that both temporal context and listing category matter when evaluating Airbnb performance.

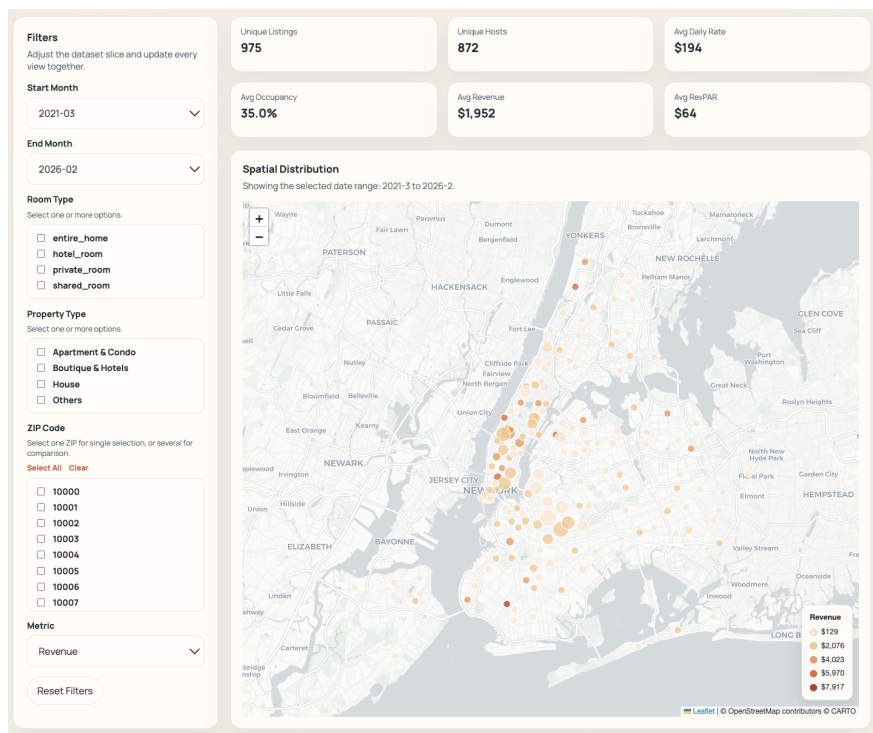


Figure 1: Spatial distribution of average Airbnb revenue across NYC ZIP codes over the selected date range. Higher-revenue areas are concentrated in a limited subset of ZIP codes rather than evenly distributed across the city.

Spatial Insight. As shown in Figure 1, higher-revenue ZIP codes are concentrated in a limited subset of areas rather than evenly distributed across New York City. This suggests that location plays a major role in earning potential and that market opportunity should be evaluated at a local submarket level rather than at the city level alone.

Temporal and Category Insight. As shown in Figure 2, Airbnb revenue varies both over time and across room types. The monthly trend indicates that market performance is not stable across the observed period, while the room type comparison shows clear differences in average revenue between accommodation formats. Together, these patterns suggest that Airbnb market performance should be interpreted through both temporal context and listing type rather than a single overall average.

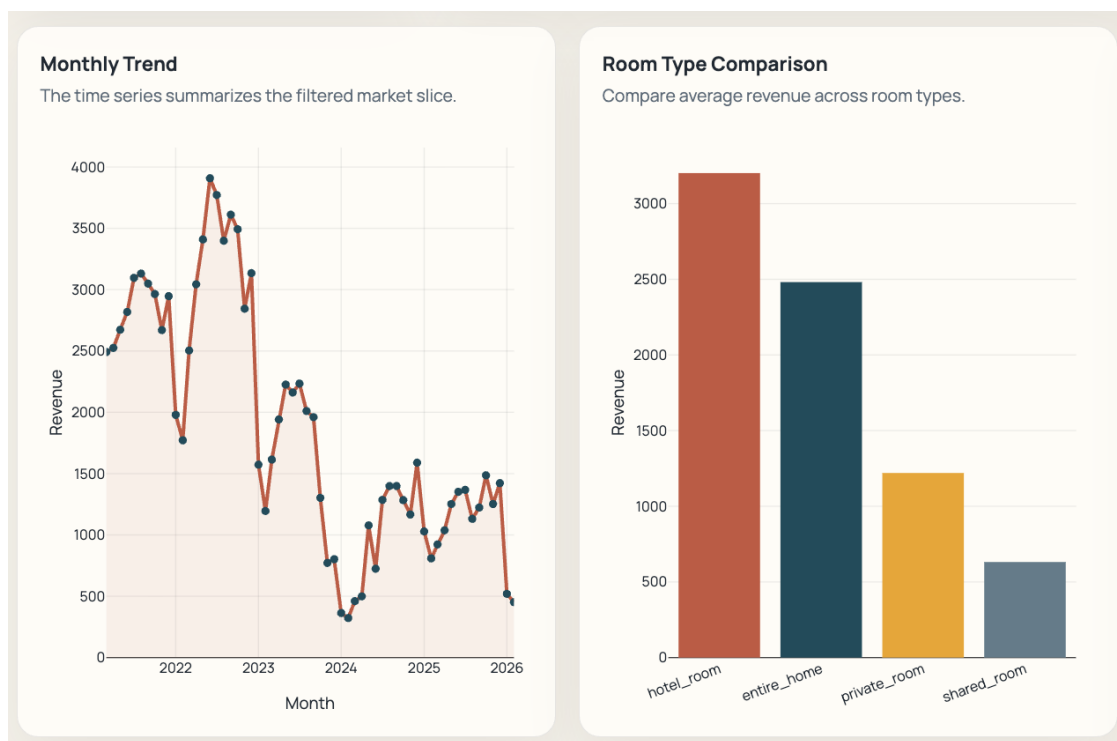


Figure 2: Monthly revenue trend and room type comparison over the selected date range, showing substantial temporal fluctuation and clear differences in average revenue across accommodation formats.

7 Limitations and Future Work

One limitation of the current system lies in the structure and quality of the source data. Although the dataset provides useful monthly Airbnb performance records, several fields are incomplete or difficult to interpret directly. For example, the original *neighborhood* field did not consistently function as a human-readable label and therefore had to be normalized into a *zip_code* field during

preprocessing. Records whose ZIP codes could not be resolved were removed from the processed dataset to avoid ambiguity in the spatial view.

In addition, many records contain *revenue* = 0 and *occupancy* = 0, which typically indicate that a listing had no bookings during a given month. This introduced an important design challenge: when visualizing revenue on the map, many ZIP codes appeared even though their average revenue was effectively zero, resulting in a cluttered and less informative spatial view. To address this, the final system hides ZIP codes with zero average revenue when the map is displayed in revenue mode. However, this also highlights a broader limitation of the dataset: monthly aggregation simplifies market behavior and may obscure finer-grained dynamics such as day-level booking patterns, short-term price adjustments, and temporal fluctuations.

A second limitation concerns the interaction design and analytical depth of the dashboard. While the system supports filtering by date range, room type, property type, ZIP code, and metric, interaction remains primarily filter-driven rather than fully coordinated through direct cross-view selection. During development, the spatial view underwent multiple revisions to improve interpretability. At one stage, the map was modified to display only the latest month within the selected date range so that changes would appear more visually distinct. However, this created a mismatch between the view’s subtitle and its actual meaning.

The final version therefore adopts a range-based aggregated map, ensuring that the selected time interval and the displayed values are semantically consistent. While this improves interpretability, it also reduces the visual variability of marker positions across longer time ranges, since ZIP-level aggregation preserves spatial structure and primarily affects color and aggregated values.

Future work could enhance the system by introducing stronger linked interactions, such as enabling users to click on a ZIP code to update all other views dynamically. Additional improvements could include providing a toggle between range-based aggregation and single-month snapshots, and integrating richer contextual data such as borough boundaries, seasonal events, or regulatory information to support deeper analysis.

8 Conclusion

This project presented an interactive web-based visualization system for exploring monthly Airbnb market dynamics in New York City. By combining spatial, temporal, and categorical views in a coordinated dashboard, the system supports analysis of how Airbnb performance varies across ZIP codes, over time, and among room types. The project also demonstrated how established visualization principles, including effective visual encoding, perceptual hierarchy, and interaction design, can be applied to a real-world urban rental dataset.

The final system contributes a practical visual analytics tool for prospective hosts, small-scale property managers, and urban market analysts. Through preprocessing, filtering, and multi-view coordination, the dashboard makes it easier to identify spatial revenue concentration, temporal fluctuation, and room-type differences that are difficult to detect in raw tables or static charts. Overall,

the project shows that an interactive visualization approach can provide a more interpretable and decision-oriented understanding of short-term rental market behavior.

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